ENGINEERING ECONOMY — FORMULA SHEET

1) Interest and Equivalence Relations

$$F = P(1+i)^n \qquad P = \frac{F}{(1+i)^n}$$

$$F = A\left(\frac{(1+i)^n - 1}{i}\right) \qquad A = F\left(\frac{i}{(1+i)^n - 1}\right)$$

$$P = A\left(\frac{(1+i)^n - 1}{i(1+i)^n}\right) \qquad A = P\left(\frac{i(1+i)^n}{(1+i)^n - 1}\right)$$

2) Present Worth and Future Worth

$$PW = \sum_{t=0}^{n} \frac{CF_t}{(1+i)^t} \qquad FW = \sum_{t=0}^{n} CF_t(1+i)^{n-t}$$

3) Annual Worth

$$AW = PW \cdot (A/P, i, n) \qquad AW = -P(A/P, i, n) + S(A/F, i, n) - A_{O\&M}$$

4) Arithmetic and Geometric Gradients

Arithmetic gradient:

$$P = G(P/G, i, n) \qquad A = G(A/G, i, n)$$

$$(P/G, i, n) = \frac{1}{i}\left[\frac{(1+i)^n - 1}{i(1+i)^n} - \frac{n}{(1+i)^n}\right]$$

$$(A/G, i, n) = \frac{1}{i} - \frac{n}{(1+i)^n - 1}$$

Geometric gradient:

$$P = A_1 \frac{1 - \left(\frac{1+g}{1+i}\right)^n}{i - g} \qquad (i \neq g)$$

5) Capitalized Cost

$$CC = \frac{A}{i} \qquad CC = P + \frac{A}{i}$$

6) Benefit–Cost

$$B/C = \frac{PW(B)}{PW(C)} \qquad \Delta(B/C) = \frac{\Delta PW(B)}{\Delta PW(C)}$$

7) Internal Rate of Return (IRR)

$$0 = \sum_{t=0}^{n} \frac{CF_t}{(1+\mathrm{IRR})^t}$$

$$0 = \sum_{t=0}^{n} \frac{\Delta CF_t}{(1+i)^t}$$

8) External Rate of Return (EROR, MIRR)

$$PW(i) = 0$$

$$FV^+ = \sum(\text{positive CF})(1+i_r)^{n-t} \qquad PV^- = \sum \frac{(\text{negative CF})}{(1+i_b)^t}$$

$$FV^+ = PV^-(1+\text{EROR})^n \qquad \text{EROR} = \left(\frac{FV^+}{PV^-}\right)^{1/n} - 1$$

9) Nominal and Effective Rates

$$i_{\text{eff}} = \left(1+\frac{r}{m}\right)^m - 1 \qquad r = m\left[(1+i_{\text{eff}})^{1/m} - 1\right]$$

10) Replacement and Economic Life

$$AW = -P(A/P,i,n) + S(A/F,i,n) - A_{O\&M} \qquad EUAC = P(A/P,i,n) - S(A/F,i,n) + A_{O\&M}$$

11) Inflation

$$(1+i_n) = (1+i_r)(1+f)$$

12) Continuous Compounding

$$F = Pe^{rn} \qquad P = Fe^{-rn}$$

13) Linear Interpolation

$$i \approx i_1 + \frac{PW_1}{PW_1 - PW_2}(i_2 - i_1)$$